Bing Fan

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Research Interests

Computer Vision, Video Understanding, and Multimedia Forensics

Education

Ph.D., Department of Computer Science and Engineering, University of North Texas, Denton, TX, USA, 2024–2027
- Advisor: Prof. Heng Fan

M.S., Cyberspace Security, Nanchang University, Nanchang, Jiangxi, China, 2024
- Advisor: Prof. Feng Ding

B.S., Software Engineering, Nanchang University, Nanchang, Jiangxi, China, 2021

Research Experience

- Research Intern, United Imaging Intelligence Boston, MA, USA, May 2026–Aug 2026
  - Research on surgical point tracking in endoscopic video
  - Won 1st place in both 2D and 3D tracking, MICCAI 2026 STIR Challenge

Honors and Awards

- 1st Place in both 2D and 3D Tracking Accuracy, STIR Challenge, MICCAI 2026 Endoscopic Vision Challenge (Team SurgTrackV2; 2D AJ 0.754, 3D AJ 0.694; official results)
- Chinese National Scholarship, 2023

Publications

Google Scholar profile: https://scholar.google.com/citations?user=pcZrxOsAAAAJ

Conference Articles

[1] **Bing Fan**, Yunhe Feng, Yan Huang, and Heng Fan, "Exploiting Textual Semantics for Robust Cross-View Object Correspondence," *Neural Information Processing Systems (**NeurIPS**)*, 2026.

[2] **Bing Fan**, Minghao Li, Hanzhi Zhang, Shaohua Dong, Naga Prudhvi Mareedu, Xiaoqiong Liu, Weishi Shi, Yunhe Feng, Yan Huang, and Heng Fan, "Towards Visual Query Segmentation in the Wild," *Neural Information Processing Systems (**NeurIPS**)*, 2026.

[3] Yue Zhou, Xinan He, Kaiqing Lin, **Bing Fan**, Feng Ding, and Bin Li, "Simplicity Prevails: The Emergence of Generalizable AIGI Detection in Visual Foundation Models," *ACM International Conference on Multimedia (**ACM MM**)*, 2026.

[4] Xin Gu, **Bing Fan**, Jiali Yao, Zhipeng Zhang, Yan Huang, Cheng Han, Heng Fan, and Libo Zhang, "Towards Long-Form Spatio-Temporal Video Grounding," *European Conference on Computer Vision (**ECCV**)*, 2026.

[5] Jiali Yao, Xinran Deng, Xin Gu, Mengrui Dai, **Bing Fan**, Zhipeng Zhang, Yan Huang, Heng Fan, and Libo Zhang, "OmniSTVG: Toward Spatio-Temporal Omni-Object Video Grounding," *International Conference on Learning Representations (**ICLR**)*, 2026.

[6] Yijun Hu*, **Bing Fan***, Xin Gu, Haiqing Ren, Dongfang Liu, Heng Fan, and Libo Zhang, "Robust Ego-Exo Correspondence with Long-Term Memory," *Neural Information Processing Systems (**NeurIPS**)*, 2025. (*equal contribution)

[7] Yue Zhou, Xinan He, Kaiqing Lin, **Bing Fan**, Feng Ding, and Bin Li, "Breaking Latent Prior Bias in Detectors for Generalizable AIGC Image Detection," *Neural Information Processing Systems (**NeurIPS**)*, 2025.

[8] Xinan He, Yue Zhou, **Bing Fan**, Bin Li, Guopu Zhu, and Feng Ding, "VLForgery Face Triad: Detection, Localization and Attribution via Multimodal Large Language Models," *Neural Information Processing Systems (**NeurIPS**)*, 2025.

[9] **Bing Fan**, Yunhe Feng, Yapeng Tian, James Chenhao Liang, Yuewei Lin, Yan Huang, and Heng Fan, "PRVQL: Progressive Knowledge-Guided Refinement for Robust Egocentric Visual Query Localization," *IEEE/CVF International Conference on Computer Vision (**ICCV**)*, 2025.

[10] **Bing Fan**, Shu Hu, and Feng Ding, "Synthesizing Black-Box Anti-Forensics DeepFakes with High Visual Quality," *IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**)*, 2024.

Journal Articles

[11] **Bing Fan**, Feng Ding, Guopu Zhu, Jiwu Huang, Sam Kwong, Pradeep Atrey, and Siwei Lyu, "Generating Higher-Quality Anti-Forensics DeepFakes with Adversarial Sharpening Mask," *ACM Transactions on Multimedia Computing, Communications and Applications (**TOMM**)*, vol. 21, 2025.

[12] Feng Ding, **Bing Fan**, Zhangyi Shen, Keping Yu, Gautam Srivastava, Kapal Dev, and Shaohua Wan, "Securing Facial Bioinformation by Eliminating Adversarial Perturbations," *IEEE Transactions on Industrial Informatics (**TII**)*, 19(5): 6682–6691, 2022.

Professional Activities

Conference Reviewers

- IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026
- Advances in Neural Information Processing Systems (**NeurIPS**), 2026
- ACM International Conference on Multimedia (**ACM MM**), 2026

Workshop Reviewers

- Workshop on Medical Video Understanding (**MedVidU**) at ECCV 2026

References

- **Prof. Heng Fan**, Assistant Professor (Ph.D. Advisor)
  Dept. of Computer Science and Engineering, University of North Texas, heng.fan@unt.edu
- **Prof. Yunhe Feng**, Assistant Professor
  Dept. of Computer Science and Engineering, University of North Texas, yunhe.feng@unt.edu
- **Prof. Yan Huang**, Regents Professor
  Dept. of Computer Science and Engineering, University of North Texas, yan.huang@unt.edu